Lecture 5

AI-CR-503

Digital Technology Program

Madan Bhandari University of Science and Technology (MBUST)

UNIT 5 Critical evaluation of solutions

Source Internet

Critical evaluation of solutions

- Critical evaluation of solutions is a process of critically analyzing the project for effectiveness, feasibility, and potential consequences before come to final decision.

- The weaknesses and strengths of solutions are measured in a structured pattern for projects.

- It asks critical questions regarding solutions.

Critical evaluation of solutions in PCAI

- It means that people-centred AI is critically evaluated is serving human needs, values, and AI performance.

- It focus on usability, trust, user autonomy with the algorithmic design and evaluation process.

- It also includes alignment of human needs, transparency, explainability, trust, and ethical risks.

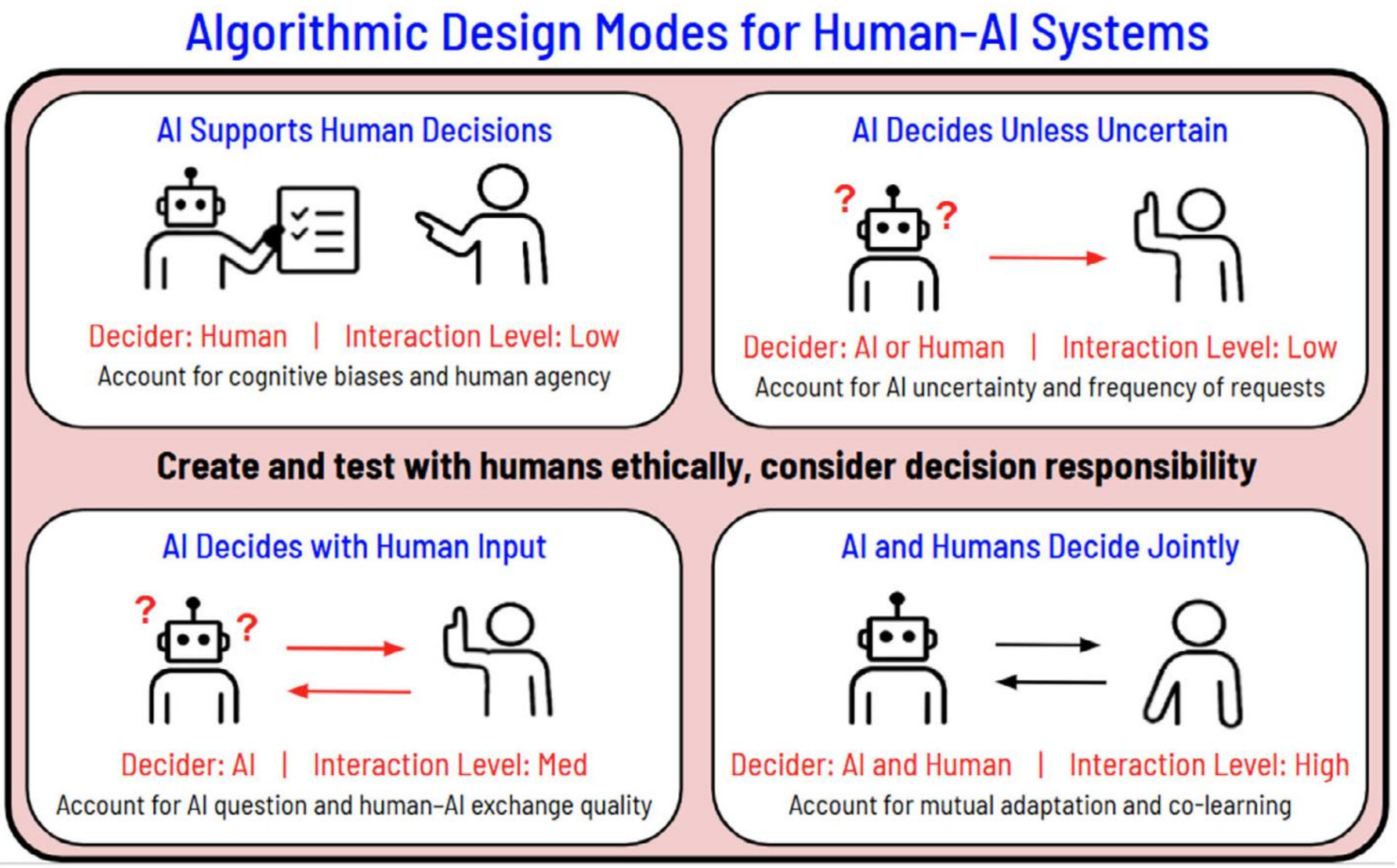
Methods of critical evaluation of solutions in PCAI

1. Algorithmic design.

2. Evaluation with real users.

3. Ethical reflection.

1. Algorithmic design

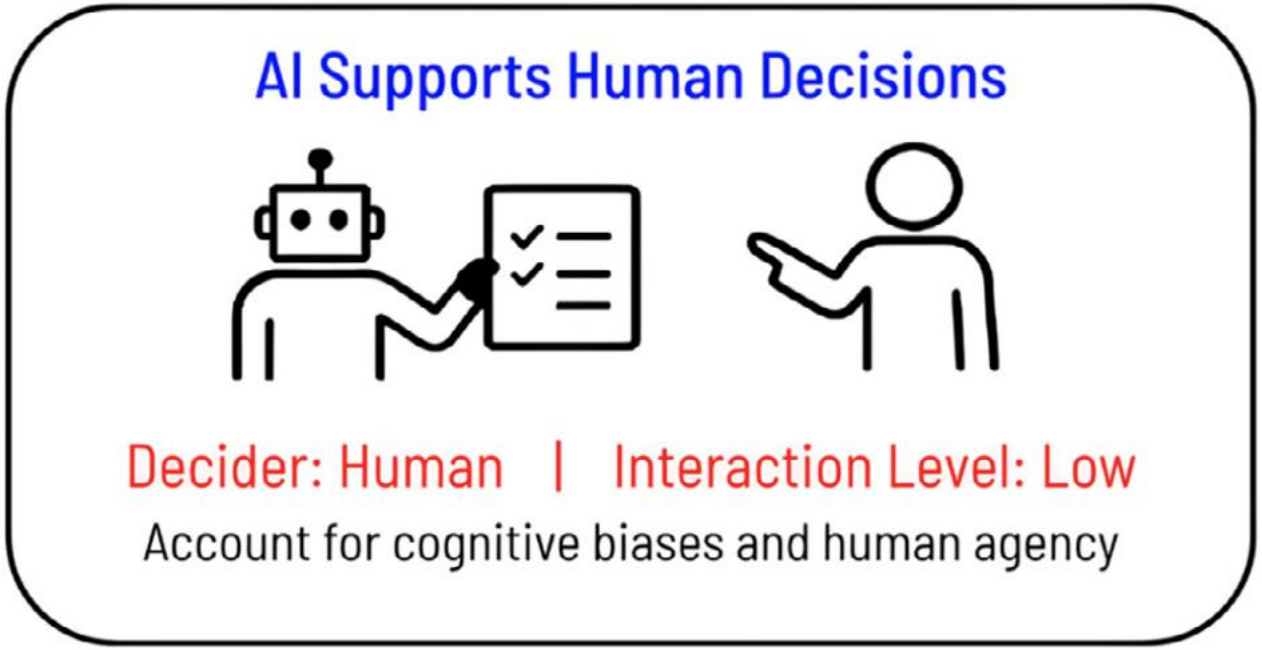


a) AI Supports Human Decisions: In this mode, AI generates predictions, recommendations, or insights to aid human decision-making, but the human is accountable for the final decision.

b) AI Decides Unless Uncertain: In this mode, AI primarily functions independently but defers to a human or abstains from making a recommendation altogether when uncertain.

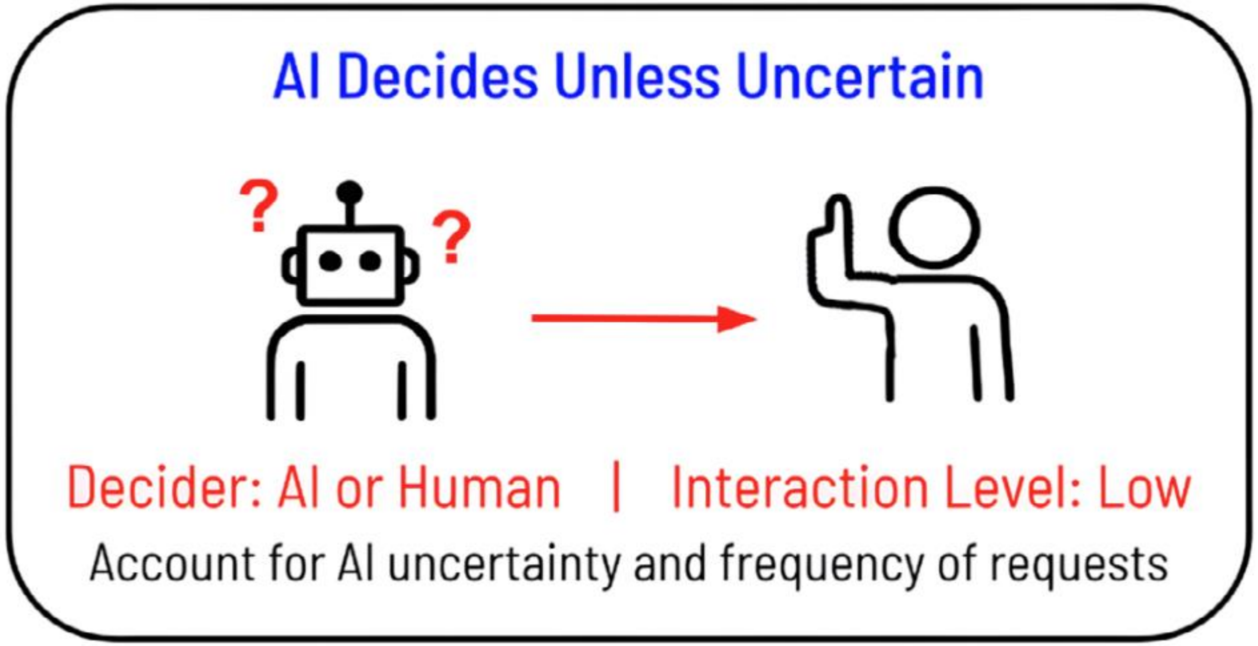
c) AI Decides with Human Input: In this mode, AI actively seeks additional input from humans or other sources to refine its decision-making when confidence is low.

d) AI and Humans Decide Jointly: In this mode, humans and AI engage in continuous, bidirectional interaction with each other and the environment in real-time without requiring explicit prompts.



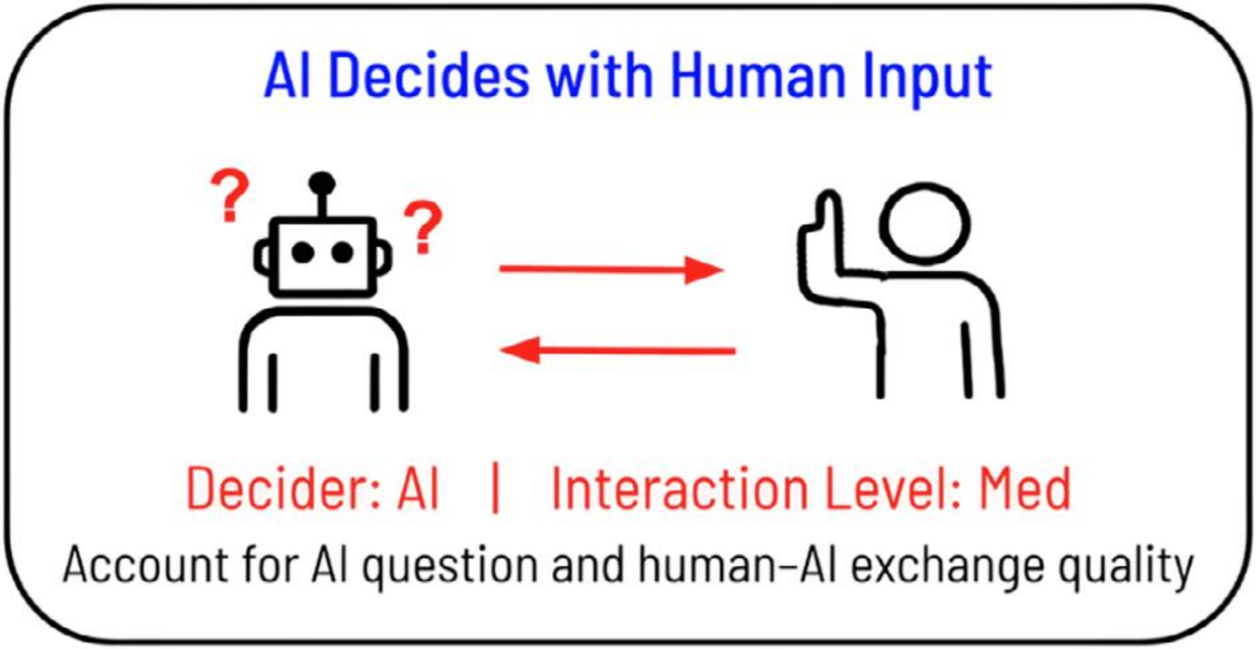
AI Supports Human Decisions
Decider: Human | Interaction Level: Low
Account for cognitive biases and human agency

1

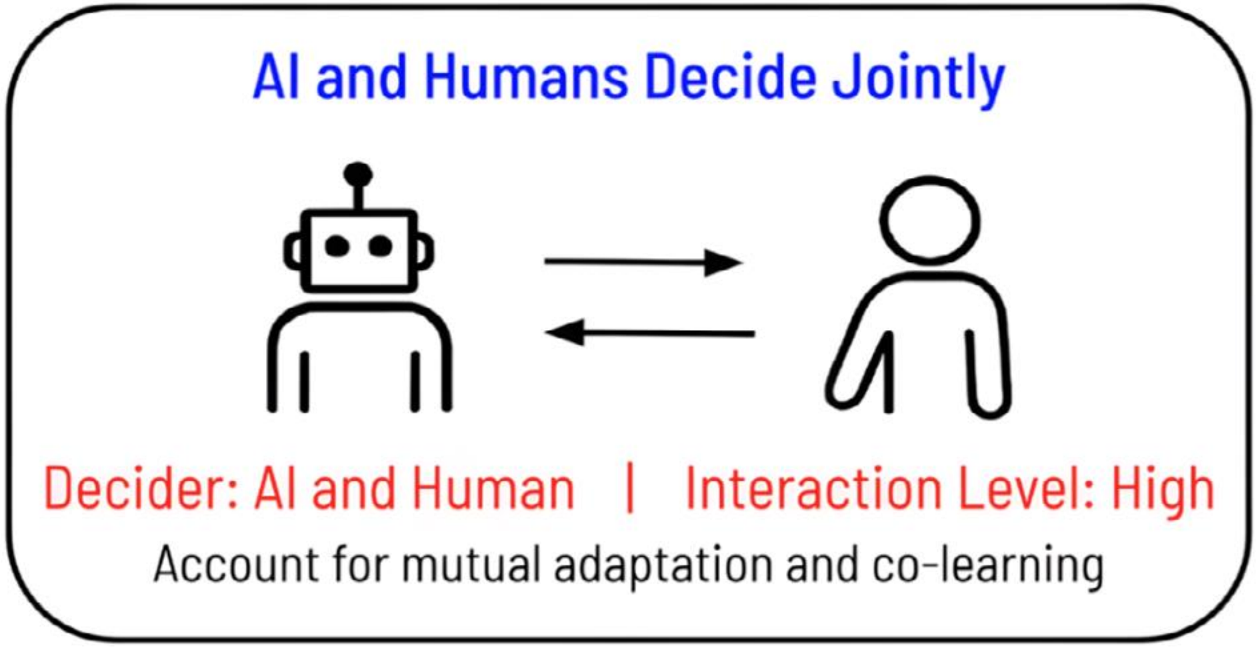


AI Decides Unless Uncertain
?
?
Decider: AI or Human | Interaction Level: Low
Account for AI uncertainty and frequency of requests

2


AI Decides with Human Input
?
?
Decider: AI | Interaction Level: Med
Account for AI question and human–AI exchange quality

3


AI and Humans Decide Jointly
Decider: AI and Human | Interaction Level: High
Account for mutual adaptation and co-learning

4

2. Evaluation with real users

- Direct evaluation with users provides invaluable insights, but it can be challenging due to costs and iteration timelines.

- Many researchers and practitioners rely on proxies such as simulations and benchmarks for evaluation.

3. Ethical reflection

- AI systems can improve performance and provide deeper insights into human AI interaction; it should be sensitive to a broad range of ethical considerations.

- Recognizing and addressing these concerns helps align AI development with societal values and underscores our broader call to build more consistent, trustworthy human–AI collaborations.

Importance of critical evaluation of AI

1. Goals: Evaluation can be marked by the type of insight sought.

- Indicator (performance, fairness, safety, robustness and reliability, behavioural features, cost).
- Distribution summarisation (aggregate, extreme, functional, manual inspection).
- Subject (system, component, algorithm).

2. Methodologies: A methodology is a collection of practices, principles, and guidelines used to conduct an evaluation.

- ➢Measurement (observations, constructs).
- ➢Task origin (operation, sample, design).
- ➢Protocol (fixed, procedural generation, adaptive, interactive).
- ➢Reference (objective, rubric, subjective, no reference).
- ➢Task mode (identification, generation).

3. The Cultures: By cultures, we refer to the people involved in the evaluation process, their norms, interests and terminology.

- Evaluators (researchers, deployers, regulators).
- Motivation (comparison, understanding, assurance).
- Discipline (AI, Psychology, Security, Economics, etc.).

Two key points

- Achieving trustworthy, effective, and ethically grounded human – AI systems requires more than improving model performance or technical sophistication.

- It demands a fundamental reorientation of the AI development process around human-centered values, goals, and contexts.

Thank you.